Atlas Mod A Modular, Low-Cost Humanoid Platform for Embodied AI and Biologically Inspired Learning

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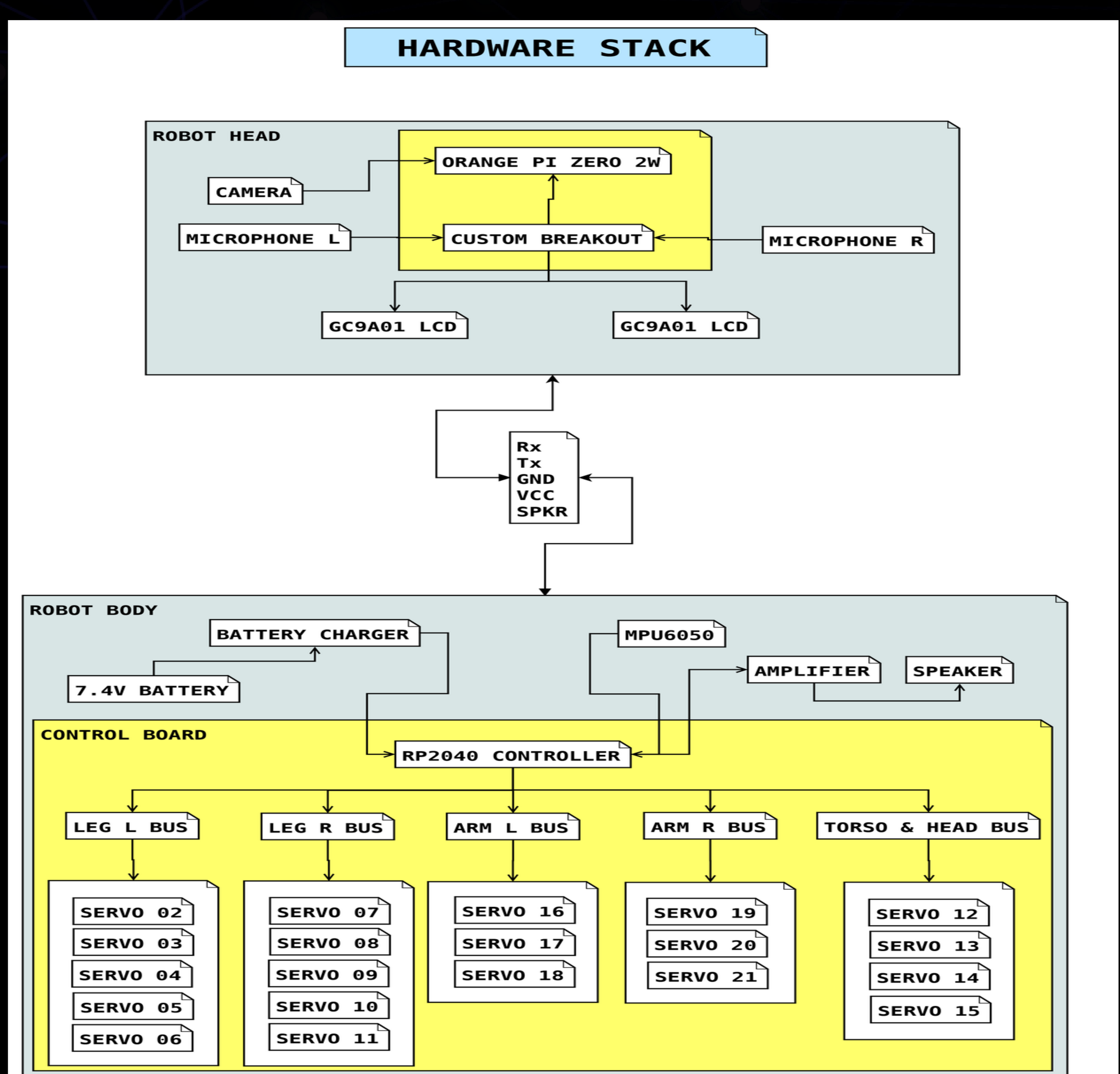
Motivation

Full-stack robot development is prohibitively expensive for independent researchers. Legacy platforms like NAO cost thousands, limiting access to institutional labs. Current consumer social robots bolt chatbots onto generic control software, lacking true environmental integration — the "AI Skyrim companion" problem.

Goal: Build a research-grade humanoid at a fraction of the cost, capable of testing machine embodiment, custom AI architectures, and genuine social interaction.

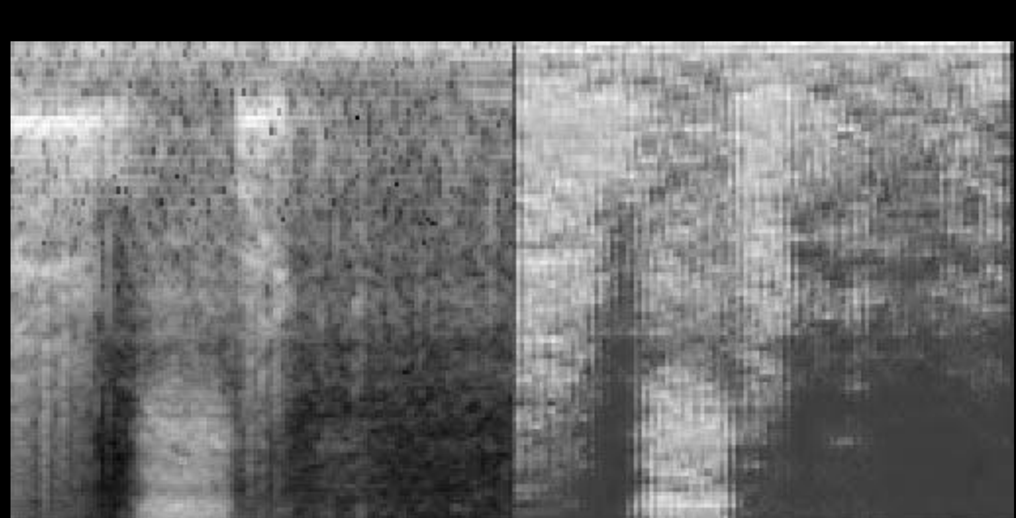
Compute Stack

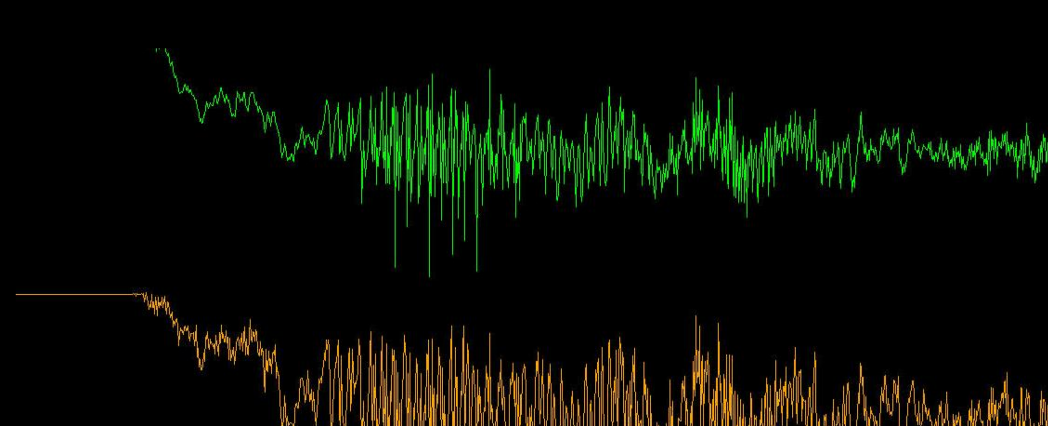
Orange Pi Zero 2W→RP2350B→Servo chains
Head: Orange Pi Zero 2W (4GB) — DietPi, AI inference
Future: Zero 3W (up to 12GB)
Body: Custom RP2350B — PIO state machines, multi-serial at 115,200 baud
Proto: RP2350 dev board + ESP32 breadboard



Neural Network Experiments

Benchmarking bio-inspired spiking networks against standard CNN baselines on captured room audio and video.
Color CNN — Baseline feature learning
Bio SNN (Color) — LIF encoder, faster convergence but parameter-sensitive
Dual Encoder CNN — Bottleneck architecture, <10 epochs
SNN Audio (encoded) — Intelligible reconstruction of "one, two, three, four"
[Triptych: Input → CNN → SNN reconstructions]
Finding: Bio SNNs learn fast when stable but easily collapse to "dead" states without careful tuning.





Hardware

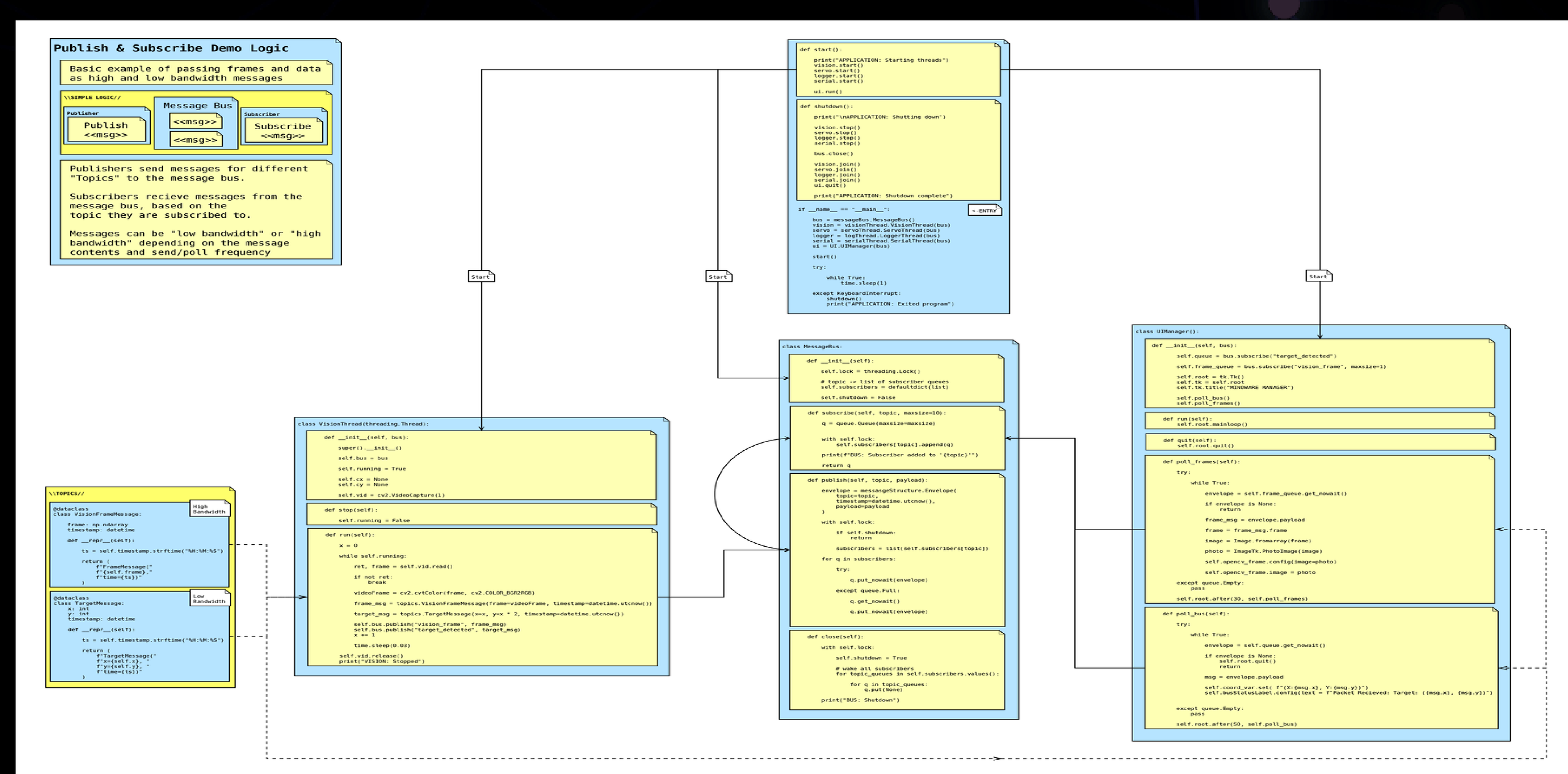
Base: De Agostini "Robi" (Takahashi, 2013) — 20× Futaba RS308MD servos, half-duplex TTL. [Photo: Chassis with custom internals]
~$600+New servo cost~$250
Used Robi units
Refurbishing skips the CAD trap

Robi Mindware

Named in homage to Sony Aibo. An integrated cognitive stack, not a chatbot bolted to a remote body.
Central bus with independent subscriber queues
Async nodes: vision, audio, IMU, motor, LLM
No main-bus congestion at high sensor frequencies

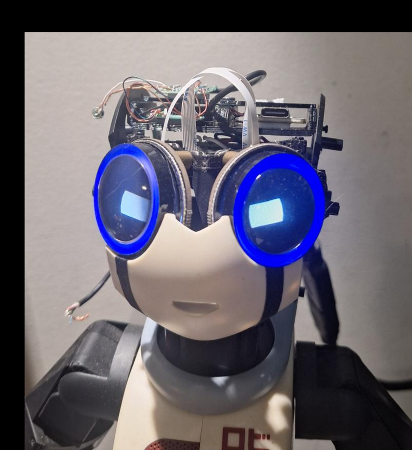
Kiosk Mode & Cloud-Edge

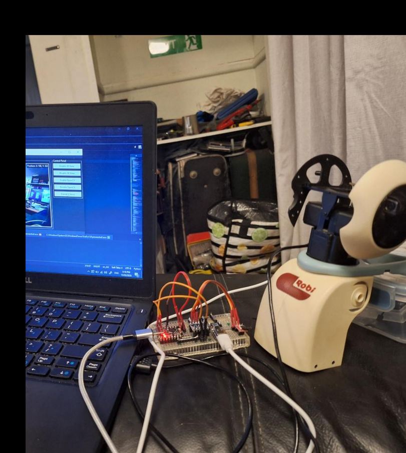
Kiosk mode runs object tracking and local voice chat via pretrained LLM. It is the functional baseline — designed to be replaced by deeper integrated AI.
Cloud-edge bifurcation: Local stack handles reflexes and sensing; cloud trains large SNNs/CNNs in real time. Brain bifurcation, not tele-operation.

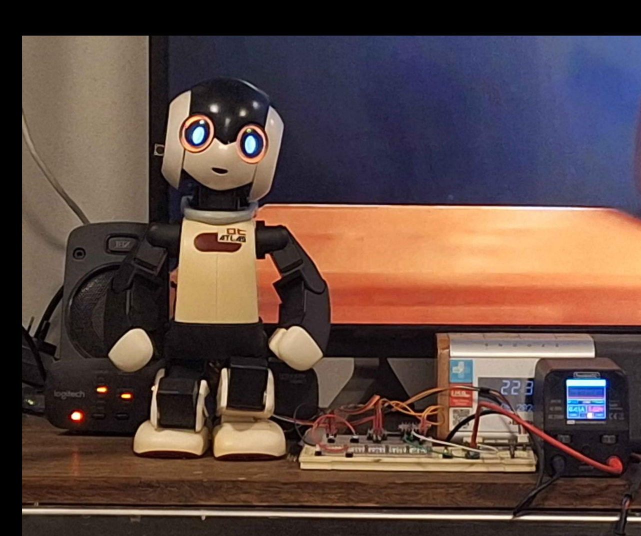


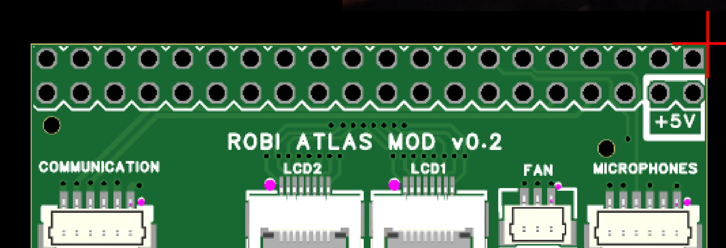
Proposed Experiments

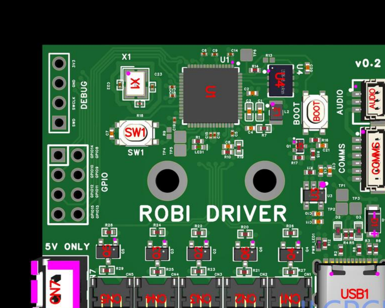
Social Learning: Two robots learn a signal, then teach it machine-to-machine without human intervention. Tests whether embodied agents can develop and transmit shared protocols. For example, teaching a besoke greeting to another robot, in which the recipient robot must learn from the teacher robot. A demonstration of knowledge transmission.











Current State

✓ Servo control validated
✓ Custom breakout board
✓ Message bus + kiosk mode live
✓ CNN / SNN experiments
✓ RP2350B schematics finalized
● Dev board verification
● PCB manufacture (JLC PCB)
● IMU + audio amp integration